Week 9 - Graded Assignment 9

The due date for submitting this assignment has passed.
Due on 2026-04-17, 23:59 IST.
You may submit any number of times before the due date. The final submission will be considered for grading.

You have last submitted on: 2026-04-17, 14:14 IST

Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.

Multiple Select Questions (MSQ)

Match the functions of two variables in Column A with their graphs given in Column B(Use the table to answer questions 1 to 4).

The graph of the function with serial number i) in column A has the serial number ______ in Column B.
2

Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 2

1 point

The graph of the function with serial number ii) in column A has the serial number ______ in Column B.

1

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 1

1 point

The graph of the function with serial number iii) in column A has the serial number ______ in Column B.

4

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 4

1 point

The graph of the function with serial number iv) in column A has the serial number ______ in Column B.

3

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 3

1 point

1 point

Consider a function $f: R^2 \rightarrow R$, such that $\frac{\partial f}{\partial x} = y\cos(xy)$. Which of the following are possible candidates for the function f?

- [x] $f(x, y) = \sin(xy)$.
- [ ] $f(x, y) = xy\cos(xy)$.
- [ ] $f(x, y) = \sin(xy) + xy$.
- [x] $f(x, y) = \sin(xy) + \cos(y)$.
- [ ] $f(x, y) = \sin(xy) + \phi(x)$, for some non-constant function $\phi(x)$.
- [x] $f(x, y) = \sin(xy) + \phi(y)$, for some non-constant function $\phi(y)$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

$f(x, y) = \sin(xy)$.

$f(x, y) = \sin(xy) + \cos(y)$.

$f(x, y) = \sin(xy) + \phi(y)$, for some non-constant function $\phi(y)$.

1 point

Consider the function $f: R^2 \rightarrow R$ defined as follows:

$$f(x, y) = \sqrt{x^2 + y^2}$$

Choose the set of correct options.

- [ ] $f_x(0,0) = 0$, and $f_y(0,0) = 0$.
- [x] $f_x(0,0)$ does not exist.
- [x] $f_y(0,0)$ does not exist.
- [x] $f_x(1,1) = \frac{1}{\sqrt{2}}$

Yes, the answer is correct.
Score: 1
Accepted Answers:
$f_x(0,0)$ does not exist.
$f_y(0,0)$ does not exist.
$f_x(1,1) = \frac{1}{\sqrt{2}}$

Numerical Answer Type (NAT):
Which of the following functions f and g from R^2 to R are continuous at the origin?

$$f(x,y) = \begin{cases} \frac{x^3}{3x^2 y} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } x = 0 \text{ or } y = 0 \end{cases}$$

$$g(x,y) = \begin{cases} x^4 + x^3 y + x y^3 + y^4 & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } x = y = 0 \end{cases}$$

- **Statement 1:** $f(x,y)$ is continuous at the origin
- **Statement 2:** $g(x,y)$ is continuous at the origin.

Find out the number of correct statements.
1
Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 1
1 point

Consider the function $f : R^2 \to R^2$ defined as follows:

$$f(x,y) = (f_1(x,y), f_2(x,y))$$

where $f_1(x,y) = xy + y^2$ and $f_2(x,y) = x + xy + 1$. Define a matrix A as follows:

$$\begin{bmatrix} \frac{\partial f_1}{\partial x}(-1,-1) & \frac{\partial f_1}{\partial y}(-1,-1) \\ \frac{\partial f_2}{\partial x}(-1,-1) & \frac{\partial f_2}{\partial y}(-1,-1) \end{bmatrix}$$

What will be the determinant of A?

1

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 1

1 point

Find out the directional derivative of the function $f(x, y) = x^2 y^3$ at $(1, 4)$, in the direction of the vector $(0, 3)$.

48

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 48

1 point

Comprehension Type Question:

The price of a product ($f(x, y)$) depends on the price (x) of the raw materials and the price (y) of transportation of the product to the market according to

$$f : R^2 \to R$$
$$f(x, y) = x^2 + xy + y^2$$

Answer Questions 14,15 and 16 using the information above.

Which of the following statements is true if the rate of change of the price of the product with respect to the price of the raw materials is the same as the rate of change of price of the product with respect to the price of transportation of the product to the market?

- **Statement 1:** The price of the raw materials is twice the price of the transportation of the product to the market.
- **Statement 2:** The price of the raw materials is half the price of the transportation of the product to the market.
- **Statement 3:** The price of the raw materials is the same as the price of the transportation of the product to the market.
- **Statement 4:** The price of the raw materials is 3 times the price of the transportation of the product to the market.

(If Statement 2 is correct, then enter 2 as your answer.)

3

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 3

1 point

1 point

Which of the following statements are true?

- [ ] $f(x, y)$ is a linear function.
- [x] $f(cx, cy) = c^2 f(x, y)$ for any real number c.
- [x] $f(x, y)$ is continuous at $(0, 0)$.
- [ ] If the price of transportation and raw material of the product approaches 0 and 5 respectively, then the price of the product approaches 30.
- [x] If the price of transportation and raw material of the product approaches 0 and 5 respectively, then the price of the product approaches 25.

Yes, the answer is correct.

Score: 1

Accepted Answers:

$f(cx, cy) = c^2 f(x, y)$ for any real number c.

$f(x, y)$ is continuous at $(0, 0)$.

If the price of transportation and raw material of the product approaches 0 and 5 respectively, then the price of the product approaches 25.

If the rate of change of the price of the product along the direction of the vector $(1, m)$ is $\frac{1}{\sqrt{1+m^2}}[ka + lb]$, when the price of raw material is a and the price of transportation of the product to the market is b, then find the value of $2k - l$.

3

Yes, the answer is correct.

Score: 1